

Probability Distributions

Full Mastery Guide

Every distribution from first principles — PMF, PDF, hypergeometric, exponential, normal — plus full worked solutions to Problem Sheets 6–10 with complete identification logic.

January 2025 · Problem Sheets 6 – 10 · MathJax typeset · Interactive charts



PMF

Probability Mass Function

The fundamental tool for discrete random variables

WHAT IS IT?

A PMF assigns a probability to every possible value of a **discrete** random variable X — one that takes countable, separated values (0, 1, 2, ...). Think of it as a lookup table: input a value x , get back $P(X = x)$. For a function to be a valid PMF it must satisfy two conditions: every probability must be non-negative, and they must all sum to exactly 1.

1 $p(x) \geq 0$ for all x

2 $\sum p(x) = 1$ (sum over all x)

3 X is discrete (countable values)

4 $p(x) = 0$ outside the support

CORE REQUIREMENTS

$$p(x) \geq 0 \quad \text{and} \quad \sum_{\text{all } x} p(x) = 1$$

Cumulative Distribution Function (CDF): $F(x) = P(X \leq x) = \sum_{t \leq x} p(t)$

$p(x) = P(X=x)$

$F(x) = P(X \leq x)$

$E[X] = \sum x \cdot p(x)$

$\text{Var}(X) = E[X^2] - (E[X])^2$

KEY

To verify a PMF: (1) check every $p(x) \geq 0$, (2) sum all $p(x)$ — must equal exactly 1. If the sum $\neq 1$ it's not a valid PMF. If any $p(x) < 0$ it's not valid. The CDF is always right-continuous and increases from 0 to 1.

PDF

Probability Density Function

The continuous analogue of the PMF

WHAT IS IT?

For a **continuous** random variable X (taking uncountably many values on an interval), we cannot assign probability to individual points — $P(X = x) = 0$ always. Instead, probability is area under the curve $f(x)$. The PDF $f(x)$ is a height; probability is the integral (area) between two values. Think of it like: $f(x) \times dx \approx P(X \text{ in tiny interval around } x)$.

- 1

f(x) ≥ 0 for all x
- 2

∫ f(x)dx = 1 (over all x)
- 3

X is continuous (interval values)
- 4

P(a≤X≤b) = ∫_a^b f(x)dx

CORE REQUIREMENTS & MOMENTS

$$f(x) \geq 0 \qquad \int_{-\infty}^{\infty} f(x) \, dx = 1$$

$$E[X] = \int_{-\infty}^{\infty} x f(x) \, dx \qquad \text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) \, dx$$

f(x) = density (not prob)

P(a≤X≤b) = area under curve

P(X=x) = 0 always

CDF: F(x) = ∫ f(t)dt

KEY

f(x) can be greater than 1! It's a density, not a probability. Only probabilities (integrals) must be ≤ 1. For a continuous variable, P(X < a) = P(X ≤ a) because the probability of hitting exactly a is zero.

B(n, p)

Binomial Distribution

Count successes in a fixed number of trials

WHY USE IT?

You run an experiment n times. Each time: success (prob p) or failure (prob 1-p). Trials are independent. You want P(exactly k successes).
Keywords: "n trials", "probability of success", "each independently".

- 1

Fixed n trials
- 2

Two outcomes (success/fail)
- 3

Constant p each trial
- 4

Independent trials

PMF & MOMENTS

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n$$

$$\text{Mean} = np \cdot \text{Variance} = np(1-p) \cdot \text{SD} = \sqrt{np(1-p)}$$

n=trials

k=successes

p=P(success)

C(n,k)=n!/(k!(n-k)!)

INTERACTIVE — B(N, P)

n

10

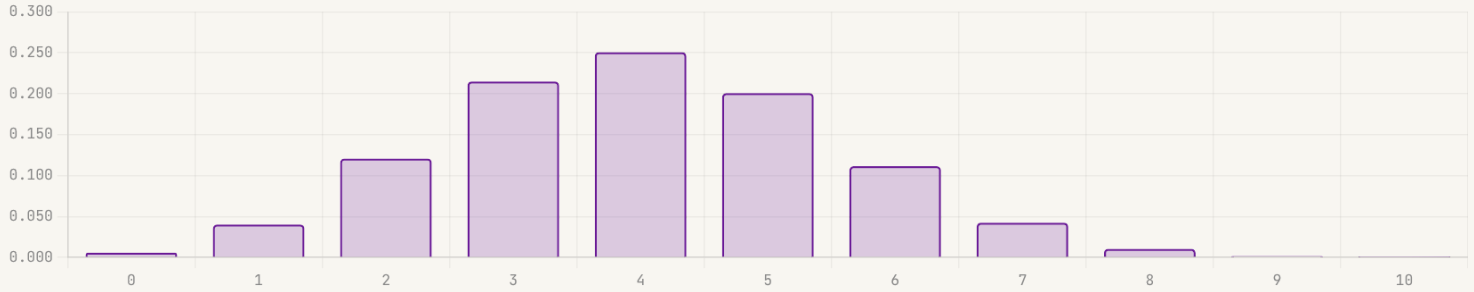
p

0.40

Mean=4.00

Var=2.40

SD=1.55



TRAP If sampling **WITHOUT** replacement from a small population, use **Hypergeometric** instead. Binomial requires independence, which is violated without replacement.

Geo(p)

Geometric Distribution

Trials until the FIRST success

WHY USE IT?

You keep repeating until you succeed for the *first* time. No fixed number of trials. Keywords: "first success", "first occurrence", "first time", "how many attempts until".

1 Independent trials

2 Constant p

3 Count until FIRST success

4 No upper limit on trials

PMF & MOMENTS

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots$$

$$\text{Mean} = 1/p \cdot \text{Variance} = (1-p)/p^2 \cdot \text{Memoryless: } P(X > m+n | X > m) = P(X > n)$$

k=trial of first success

p=P(success)

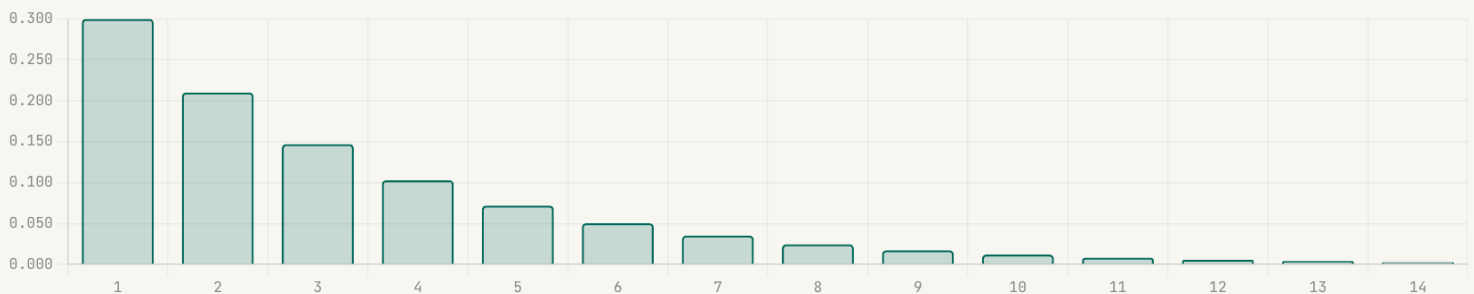
Mean=1/p

INTERACTIVE - GEO(P)

p

Mean=3.33

Var=7.78



NB(r, p)

Negative Binomial

Trials to reach the r-th success

WHY USE IT?

Generalises the geometric: wait for the r -th success instead of just the first. The key insight — the very last trial *must* be a success, so we arrange the $r-1$ prior successes among the first $n-1$ trials. Keywords: "r-th success", "collect r coupons", "r failures before stopping".

1 Independent trials

2 Constant p

3 Target r successes

4 Count total trials taken

PMF — R-TH SUCCESS ON TRIAL N

$$P(X = n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}, \quad n = r, r+1, \dots$$

Mean = r/p · Variance = $r(1-p)/p^2$ · When $r=1$, reduces to Geometric

r=target successes

n=total trials

p=P(success)

Mean=r/p

Po(λ)

Poisson Distribution

Events in a continuous interval at a constant rate

WHY USE IT?

Count rare events over a fixed time, area, or distance when you know the average rate λ but there's no natural "n trials". Keywords: "per hour", "per km", "average rate", "events in an interval". Mean = Variance = λ is the signature.

1 Events are independent

2 Constant average rate λ

3 Fixed interval

4 No two events simultaneously

PMF & SCALING

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Mean = Variance = λ · If rate is λ /hour and you want λ for t hours: $\lambda_{\text{new}} = \lambda \cdot t$

λ =avg events/interval

k=count of events

$e \approx 2.71828$

Scale: $\lambda \cdot t = \lambda \cdot t$

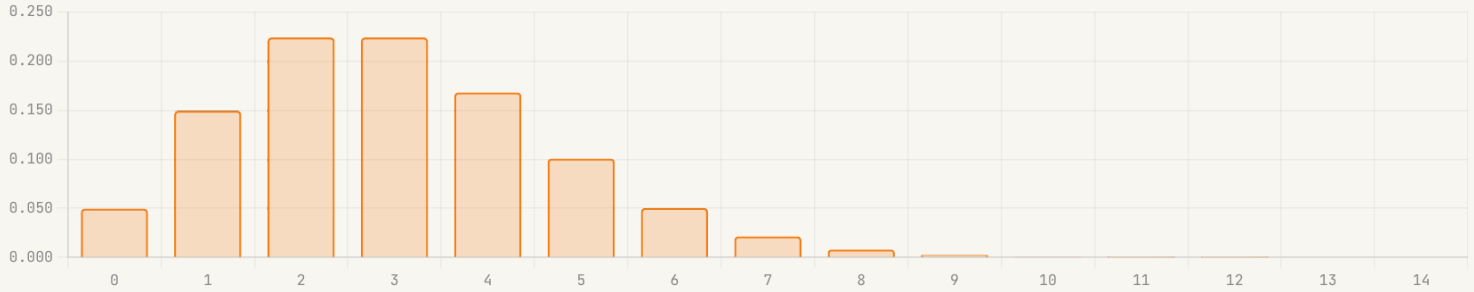
INTERACTIVE — PO(λ)

λ

Mean=3

Var=3

SD=1.73



$H(N, K, n)$

Hypergeometric Distribution

Sampling *WITHOUT* replacement from a finite population

WHY USE IT?

You have a finite population of N objects: K are "successes" and $N-K$ are "failures". You draw n objects **without replacement**. Since each draw changes the remaining population, draws are NOT independent — so Binomial fails. The hypergeometric handles this exactly. Keywords: "from an urn", "without replacement", "finite population", "selected from a group".

1 Finite population of N objects

2 K successes in population

3 Draw n objects

4 **WITHOUT** replacement (dependent)

PMF & MOMENTS

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

Choose k from K successes $\times$ choose remaining $(n-k)$ from $N-K$ failures, divided by all ways to choose n from N

$$\text{Mean} = n \cdot \frac{K}{N} \quad \text{Var} = n \cdot \frac{K}{N} \cdot \frac{N-K}{N} \cdot \frac{N-n}{N-1}$$

N =population size

K =successes in pop.

n =sample size

k =successes in sample

$(N-n)/(N-1)$ = finite pop. correction

KEY Hypergeometric vs Binomial: if sampling **WITH** replacement (or population is huge), use Binomial. If sampling **WITHOUT** replacement from a small finite population, use Hypergeometric. The formula counts combinations directly — how many ways to pick k successes from K , times the ways to pick the remaining $n-k$ from the $N-K$ failures.

$\text{Exp}(\lambda)$

Exponential Distribution

Waiting time between Poisson events — continuous, memoryless

WHY USE IT?

If events arrive at a Poisson rate of λ per unit time, then the *time between consecutive events* follows $\text{Exp}(\lambda)$. It models lifetimes, waiting times, and inter-arrival times. Like the geometric, it is memoryless: having waited t minutes gives you no advantage for the future. Keywords: "time until", "lifetime", "waiting time", "inter-arrival time".

1 Continuous waiting time $X \geq 0$

2 Events occur at constant Poisson rate λ

3 Memoryless property holds

4 Only one type of event

PDF, CDF & MOMENTS

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}$$

$$\text{Mean} = \frac{1}{\lambda} \quad \text{Variance} = \frac{1}{\lambda^2} \quad \text{SD} = \frac{1}{\lambda}$$

λ =rate (events/time)

Mean= $1/\lambda$

$P(X>t)=e^{-\lambda t}$

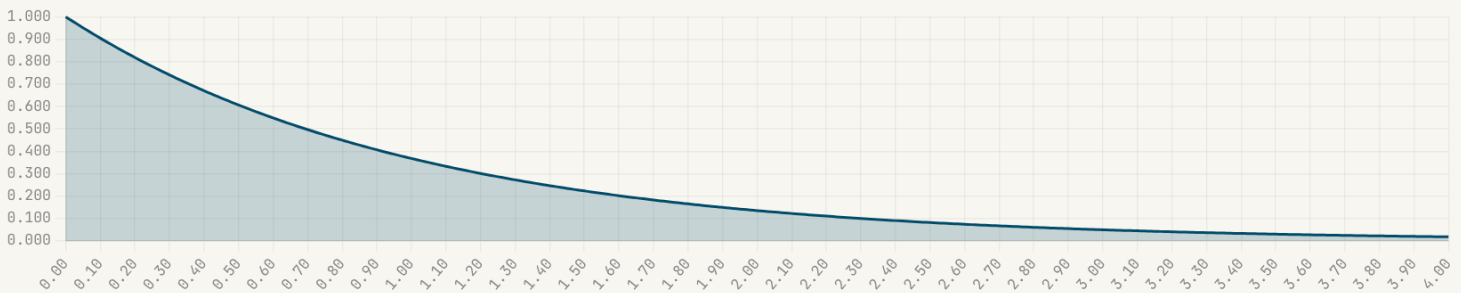
Memoryless: $P(X>s+t|X>s)=P(X>t)$

INTERACTIVE — EXP(λ) PDF

λ

Mean=1.00

Var=1.00



$N(\mu, \sigma^2)$

Normal Distribution

The bell curve — the most important distribution in statistics

WHY USE IT?

By the Central Limit Theorem, means of large samples from ANY distribution are approximately normal. It models natural continuous variation: heights, exam scores, measurement errors. Key: always convert to z-score = $(x-\mu)/\sigma$ to use standard normal tables. Keywords: "normally distributed", "mean and variance given", "large sample mean".

PDF, Z-SCORE & EMPIRICAL RULE

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

68.3%

95.4%

99.7%

within $\mu \pm \sigma$

within $\mu \pm 2\sigma$

within $\mu \pm 3\sigma$

μ =mean

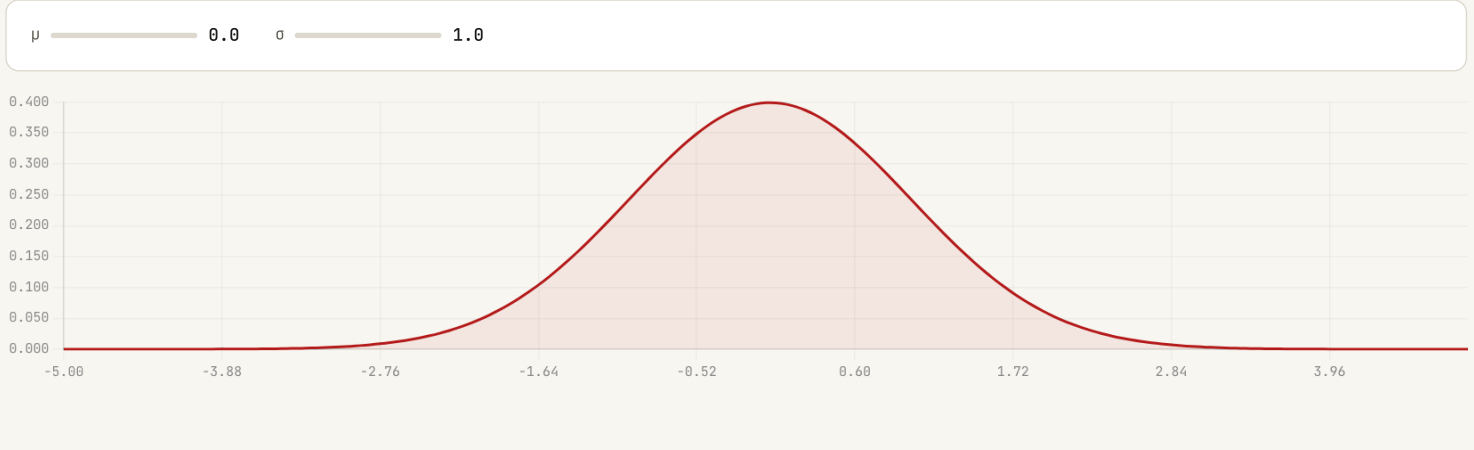
σ =std dev

σ^2 =variance

$z=(x-\mu)/\sigma$

$P(X=x)=0$ always

INTERACTIVE — $N(\mu, \sigma^2)$



? How to Identify the Distribution

A systematic decision process — ask these questions in order

Step 1 — Is X discrete or continuous?

Discrete X takes countable values (0,1,2,...). You're counting something. → Go to Step 2

Continuous X takes any value in an interval (time, weight, height). → Jump to Step 5

Step 2 — Are you counting successes or waiting?

Counting successes You want to count how many successes occur. → Go to Step 3

Waiting for success You want to know how many trials until success(es). → Go to Step 4

Counting events in interval No fixed trials, just events in a window at some rate. → **Poisson**

Sampling without replacement Finite population, draws are dependent. → **Hypergeometric**

Step 3 — Is n fixed?

Yes, fixed n Fixed number of independent trials, constant p . → **Binomial $B(n,p)$**

No fixed n Unbounded number of trials. Are you waiting for events? → Step 4

Step 4 — Waiting for how many successes?

First success only How many trials until the 1st success? → **Geometric $Geo(p)$**

r-th success ($r > 1$) How many trials until the r-th success? → **Negative Binomial $NB(r,p)$**

Step 5 — Continuous: what shape?

Waiting time / lifetime Time until next event, linked to Poisson process, memoryless. → **Exponential $Exp(\lambda)$**

Symmetric bell shape Natural variation, mean $\pm$ SD, z-scores. → **Normal $N(\mu,\sigma^2)$**

Problem Sheet 6 — PMF Verification & CDF

Testing whether functions are valid PMFs, finding constants, plotting, and deriving PMF from CDF.

Q1(a) Is $p(x) = 0.25 \cdot (0.75)^x$ for $x = 0, 1, 2, \dots$ a valid PMF?

STEP 1 – IDENTIFY THE STRUCTURE

Read the function carefully. We have $p(x) = 0.25 \cdot (0.75)^x$.

SYMBOL	VALUE	ROLE
0.25	first factor	This is $p(0)$ — the probability when $x=0$
$(0.75)^x$	decaying factor	Makes probabilities decrease as x grows
x	0, 1, 2, 3, ...	Countably infinite support

This looks like a **Geometric-type PMF**. In a Geometric with success probability p , we have $P(X = x) = (1 - p)^x \cdot p$ for $x = 0, 1, 2, \dots$. Here $p = 0.25$ and $(1-p) = 0.75$. ✓

STEP 2 – CHECK CONDITION 1: $P(X) \geq 0$

Is $0.25 \cdot (0.75)^x \geq 0$ for all $x = 0, 1, 2, \dots$?

$0.25 > 0$ ✓ and $(0.75)^x > 0$ for all x ✓. So yes, every $p(x)$ is strictly positive. Condition 1 satisfied.

STEP 3 – CHECK CONDITION 2: SUM = 1

We must sum over all $x = 0, 1, 2, \dots$:

$$\sum_{x=0}^{\infty} 0.25 \cdot (0.75)^x = 0.25 \sum_{x=0}^{\infty} (0.75)^x$$

This is a geometric series $\sum_{i=0}^{\infty} a^i = \frac{1}{1-a}$ where $a = 0.75$, so:

$$= 0.25 \cdot \frac{1}{1-0.75} = 0.25 \cdot \frac{1}{0.25} = 0.25 \cdot 4 = 1$$

CONCLUSION

✓ **Valid PMF – both conditions satisfied. This is Geometric($p=0.25$).**

Q1(b) Is $p(x) = \frac{1}{5}(2x - 3)$ for $x = 0, 1, 2, 3, 4, 5$ a valid PMF?

STEP 1 – COMPUTE $P(X)$ FOR EACH X

x	$2x-3$	$P(X) = (2x-3)/5$	VALID?
0	-3	$-3/5 = -0.6$	✗ NEGATIVE
1	-1	$-1/5 = -0.2$	✗ NEGATIVE
2	1	$1/5 = 0.2$	✓

3	3	$3/5 = 0.6$	✓
4	5	$5/5 = 1.0$	✓
5	7	$7/5 = 1.4$	✗ > 1

CONCLUSION

✗ NOT a valid PMF – $p(0)$ and $p(1)$ are negative, violating $p(x) \geq 0$.

Q1(c) Is $p(x) = \frac{\binom{5}{x}\binom{4}{x-2}}{36}$ for $x = 0, 1, 2$ a valid PMF?

STEP 1 – IDENTIFY THE STRUCTURE

This looks like a **Hypergeometric**-type formula. We choose x items from one group of 5, and $x-2$ items from another group of 4. Let's evaluate each x .

Recall: $\binom{n}{k} = 0$ if $k < 0$ or $k > n$.

STEP 2 – COMPUTE EACH VALUE

X	$C(5, X)$	$C(4, X-2)$	PRODUCT	$P(X) = /36$
0	$C(5,0)=1$	$C(4,-2)=0$	0	$0/36 = 0$
1	$C(5,1)=5$	$C(4,-1)=0$	0	$0/36 = 0$
2	$C(5,2)=10$	$C(4,0)=1$	10	$10/36 \approx 0.278$

Sum check: $0 + 0 + 10/36 = 10/36 \neq 1$. The sum only reaches $10/36 \approx 0.278$.

CONCLUSION

✗ NOT a valid PMF – probabilities only sum to $10/36$, not 1.

Q2 Find k so that $p(x) = \frac{k}{x!}$ for $x=0,1,2,3,4$ is a valid PMF. Then find $P(X < 2)$.

STEP 1 – IDENTIFY THE PATTERN

The form $\frac{1}{x!}$ appears in the Poisson PMF $\frac{\lambda^x e^{-\lambda}}{x!}$. If we ignore the $\lambda^x e^{-\lambda}$ part, we just have $\frac{1}{x!}$. This is a truncated Poisson at $\lambda=1$ — but we don't need to know that. We just use the sum-to-1 condition.

STEP 2 – COMPUTE EACH $P(X)$ AND SET SUM = 1

X	$X!$	$k/X!$
0	$0! = 1$	$k/1 = k$
1	$1! = 1$	$k/1 = k$
2	$2! = 2$	$k/2$
3	$3! = 6$	$k/6$
4	$4! = 24$	$k/24$

$$\text{Sum} = k \left(1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \right) = k \cdot \frac{24 + 24 + 12 + 4 + 1}{24} = k \cdot \frac{65}{24} = 1$$

$$k = \frac{24}{65} \approx 0.3692$$

STEP 3 – FIND $P(X < 2) = P(X=0) + P(X=1)$

$$P(X < 2) = \frac{k}{0!} + \frac{k}{1!} = k + k = 2k = \frac{48}{65} \approx 0.738$$

ANSWERS

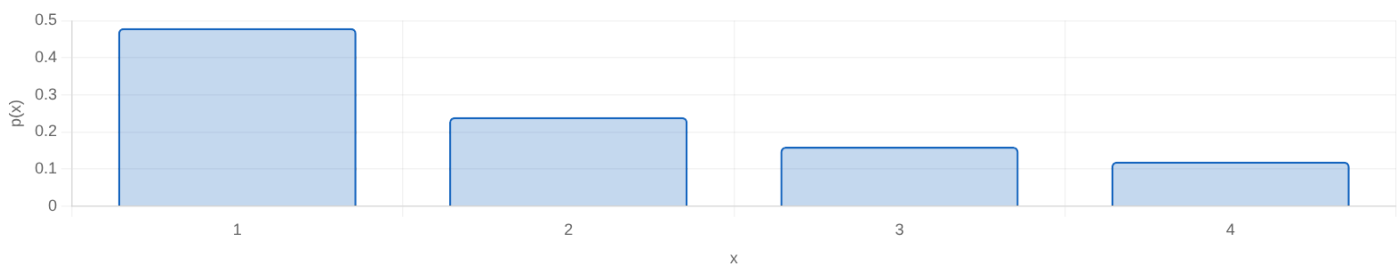
$$k = 24/65 \approx 0.369 \cdot P(X < 2) = 48/65 \approx 0.738$$

Q3 PMF: $p(x) = \frac{12}{25x}$ for $x=1,2,3,4$. Plot the bar graph.

STEP 1 – COMPUTE EACH $P(X)$

x	CALCULATION	P(X)
1	$12/(25 \times 1)$	$12/25 = 0.480$
2	$12/(25 \times 2)$	$12/50 = 0.240$
3	$12/(25 \times 3)$	$12/75 = 0.160$
4	$12/(25 \times 4)$	$12/100 = 0.120$

Sum check: $0.480 + 0.240 + 0.160 + 0.120 = 1.000$ ✓



NOTE

The distribution is right-skewed – probability decreases as x increases. This is valid since all $p(x) \geq 0$ and they sum to 1.

Q4 Given CDF $F(x)$, find the PMF of X .

STEP 1 – UNDERSTAND THE LINK: PMF FROM CDF

The CDF $F(x) = P(X \leq x)$. To go from CDF $\rightarrow$ PMF, find where $F(x)$ jumps. Each jump is a point mass:

$$p(x) = P(X = x) = F(x) - F(x^-) = F(x) - \lim_{t \rightarrow x^-} F(t)$$

In practice for discrete: $p(x) = F(x) - F(x-1)$ at each integer x .

STEP 2 – COMPUTE JUMPS

x	F(X)	F(X-1)	P(X) = F(X) - F(X-1)
0	1/2	F(-1)=0	$1/2 - 0 = 1/2$

1	3/5	1/2	3/5 - 1/2 = 6/10 - 5/10 = 1/10
2	4/5	3/5	4/5 - 3/5 = 1/5
3	9/10	4/5	9/10 - 8/10 = 1/10
4	1	9/10	1 - 9/10 = 1/10

Sum check: 5/10 + 1/10 + 2/10 + 1/10 + 1/10 = 10/10 = 1 ✓

PMF ANSWER

p(0)=1/2 · p(1)=1/10 · p(2)=1/5 · p(3)=1/10 · p(4)=1/10

Problem Sheet 7 — E[X] and Variance

Finding expected values and variances from PDFs and PMFs; working with continuous random variables.

Q1 f(x) = 2x for 0≤x≤1. Find α such that P(X < α) = 3·P(X ≥ α).

DISTRIBUTION IDENTIFIED

Continuous — use PDF integration

X is continuous (takes all values in [0,1]). We use ∫ f(x)dx to find probabilities. No discrete distribution applies here.

STEP 1 – VERIFY F(X) IS A VALID PDF

$$\int_0^1 2x \, dx = [x^2]_0^1 = 1 - 0 = 1 \checkmark$$

STEP 2 – SET UP THE EQUATION

Since P(X < α) + P(X ≥ α) = 1, and P(X < α) = 3·P(X ≥ α):

$$3P(X \geq \alpha) + P(X \geq \alpha) = 1 \implies P(X \geq \alpha) = \frac{1}{4}$$

STEP 3 – SOLVE VIA INTEGRATION

$$P(X \geq \alpha) = \int_{\alpha}^1 2x \, dx = [x^2]_{\alpha}^1 = 1 - \alpha^2 = \frac{1}{4}$$

$$\alpha^2 = \frac{3}{4} \implies \alpha = \frac{\sqrt{3}}{2} \approx 0.866$$

ANSWER

α = √3/2 ≈ 0.866

Q2 f(x) = e^{-x} for x≥0. Find t such that P(X < t) = 0.5.

DISTRIBUTION IDENTIFIED

Exponential Exp($\lambda=1$)

$f(x) = e^{-x} = 1 \cdot e^{-1x}$ is exactly the exponential PDF with $\lambda=1$. This models waiting times. Mean = $1/\lambda = 1$.

STEP 1 - SET UP THE INTEGRAL

$$P(X < t) = \int_0^t e^{-x} dx = [-e^{-x}]_0^t = 1 - e^{-t} = 0.5$$

STEP 2 - SOLVE FOR T

$$e^{-t} = 0.5 \implies -t = \ln(0.5) \implies t = -\ln(0.5) = \ln(2) \approx 0.693$$

ANSWER

$$t = \ln(2) \approx 0.693$$

Q3 Softball player: $P(X=1)=0.3$, $P(X=2)=0.2$, $P(X=0)=3P(X=3)$. Find $E(X)$ and $V(X)$.

DISTRIBUTION IDENTIFIED

General Discrete — find PMF from constraints

X is discrete (hits = 0,1,2,3). No standard named distribution is specified. We must find $p(3)$ using the sum-to-1 constraint, then compute moments directly.

STEP 1 - FIND UNKNOWN PROBABILITY

Let $P(X=3) = p$. Then $P(X=0) = 3p$. All probabilities sum to 1:

$$3p + 0.3 + 0.2 + p = 1 \implies 4p = 0.5 \implies p = 0.125$$

x	$P(X=x)$
0	$3 \times 0.125 = 0.375$
1	0.300
2	0.200
3	0.125

STEP 2 - COMPUTE $E[X]$

$$E[X] = \sum x \cdot p(x) = 0(0.375) + 1(0.3) + 2(0.2) + 3(0.125) = 0 + 0.3 + 0.4 + 0.375 = 1.075$$

STEP 3 - COMPUTE $\text{Var}(X)$ USING $E[X^2] - (E[X])^2$

$$E[X^2] = 0^2(0.375) + 1^2(0.3) + 2^2(0.2) + 3^2(0.125) = 0 + 0.3 + 0.8 + 1.125 = 2.225$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 2.225 - (1.075)^2 = 2.225 - 1.1556 = 1.069$$

ANSWERS

$E[X] = 1.075 \text{ hits} \cdot \text{Var}(X) \approx 1.069$

Q5 $f(x) = \frac{3}{4}(kx - x^2)$ for $0 \leq x \leq 2$. Find k , $E[X]$ and $\text{Var}(X)$.

STEP 1 - FIND k USING $\int f(x) dx = 1$

$$\int_0^2 \frac{3}{4}(kx - x^2) dx = 1$$

$$\frac{3}{4} \left[k \cdot \frac{x^2}{2} - \frac{x^3}{3} \right]_0^2 = \frac{3}{4} \left(2k - \frac{8}{3} \right) = 1$$

$$2k - \frac{8}{3} = \frac{4}{3} \implies 2k = 4 \implies k = 2$$

STEP 2 - FIND $E[X]$ WITH $k=2$: $f(x) = \frac{3}{4}(2x - x^2)$

$$E[X] = \int_0^2 x \cdot \frac{3}{4}(2x - x^2) dx = \frac{3}{4} \int_0^2 (2x^2 - x^3) dx$$

$$= \frac{3}{4} \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^2 = \frac{3}{4} \left(\frac{16}{3} - 4 \right) = \frac{3}{4} \cdot \frac{4}{3} = 1$$

STEP 3 - FIND $E[X^2]$ THEN $\text{Var}(X)$

$$E[X^2] = \frac{3}{4} \int_0^2 (2x^3 - x^4) dx = \frac{3}{4} \left[\frac{x^4}{2} - \frac{x^5}{5} \right]_0^2 = \frac{3}{4} \left(8 - \frac{32}{5} \right) = \frac{3}{4} \cdot \frac{8}{5} = \frac{6}{5}$$

$$\text{Var}(X) = \frac{6}{5} - 1^2 = \frac{1}{5} = 0.2$$

ANSWERS

$k = 2 \cdot E[X] = 1 \cdot \text{Var}(X) = 1/5 = 0.2$

Problem Sheet 8 — Binomial & Poisson Applications

Identifying and applying the correct named discrete distribution from context clues in the problem.

Q1 True/False test, 5 questions. Pass = at least 4 correct. If guessing, $P(\text{pass})?$

DISTRIBUTION IDENTIFIED - BINOMIAL $B(5, 0.5)$

Why Binomial?

- ✓ Fixed $n = 5$ trials (questions)
- ✓ Two outcomes per question: correct (success) or wrong (failure)
- ✓ $p = 0.5$ per question (guessing T/F randomly = 50%)
- ✓ Questions are independent (guessing one doesn't affect another)
 - All 4 Binomial conditions met. $X \sim B(5, 0.5)$

STEP 1 - IDENTIFY VARIABLES

VARIABLE	VALUE	WHY
n	5	5 questions = 5 trials
p	0.5	Random T/F guess = 50% chance correct
X	number correct	This is what we're counting
Pass event	$X \geq 4$	Need at least 4 correct

STEP 2 - COMPUTE $P(X=4)$ AND $P(X=5)$

$$P(X = 4) = \binom{5}{4} (0.5)^4 (0.5)^1 = 5 \times \frac{1}{32} = \frac{5}{32}$$

$$P(X = 5) = \binom{5}{5} (0.5)^5 = 1 \times \frac{1}{32} = \frac{1}{32}$$

$$P(\text{pass}) = P(X \geq 4) = \frac{5}{32} + \frac{1}{32} = \frac{6}{32} = \frac{3}{16} \approx 0.1875$$

ANSWER

$P(\text{pass}) = 3/16 = 0.1875$ (about 18.75%)

Q2 Engines fail independently with $P(\text{fail})=0.1$. Successful if at most half fail. $P(\text{success})$ for 2-engine and 4-engine planes?

DISTRIBUTION - BINOMIAL $B(N, 0.1)$

Why Binomial?

Each engine is a trial: it either fails ($p=0.1$) or doesn't (0.9). Engines operate independently. We count how many fail. → $X \sim B(n, 0.1)$ where $n=2$ or $n=4$.

2-ENGINE PLANE ($N=2$): SUCCESS IF AT MOST 1 FAILS → $X \leq 1$

$$P(X = 0) = (0.9)^2 = 0.81 \quad P(X = 1) = \binom{2}{1} (0.1)(0.9) = 0.18$$

$$P(\text{success}) = P(X \leq 1) = 0.81 + 0.18 = 0.99$$

4-ENGINE PLANE ($N=4$): SUCCESS IF AT MOST 2 FAIL → $X \leq 2$

$$P(X = 0) = (0.9)^4 = 0.6561$$

$$P(X = 1) = \binom{4}{1}(0.1)(0.9)^3 = 4(0.1)(0.729) = 0.2916$$

$$P(X = 2) = \binom{4}{2}(0.1)^2(0.9)^2 = 6(0.01)(0.81) = 0.0486$$

$$P(\text{success}) = 0.6561 + 0.2916 + 0.0486 = 0.9963$$

ANSWERS

2-engine: $P(\text{success}) = 0.99$ · 4-engine: $P(\text{success}) = 0.9963$

Q3 30% of 15 cars want to turn right. Lane holds 3. P(lane insufficient)?

DISTRIBUTION – BINOMIAL $B(15, 0.3)$

Why Binomial?

$n=15$ fixed cars. Each car either wants to turn right ($p=0.3$) or not. Cars decide independently. We count how many want to turn right. → $X \sim B(15, 0.3)$.
Lane insufficient when $X > 3$, i.e., $P(X \geq 4)$.

STEP 1 – COMPLEMENT: $P(X \geq 4) = 1 - P(X \leq 3)$

$$P(X = 0) = \binom{15}{0}(0.3)^0(0.7)^{15} = 0.7^{15} \approx 0.00475$$

$$P(X = 1) = 15(0.3)(0.7)^{14} \approx 15(0.3)(0.00678) = 0.03052$$

$$P(X = 2) = \binom{15}{2}(0.3)^2(0.7)^{13} \approx 105(0.09)(0.00969) = 0.09156$$

$$P(X = 3) = \binom{15}{3}(0.3)^3(0.7)^{12} \approx 455(0.027)(0.01384) = 0.17004$$

$$P(X \leq 3) \approx 0.00475 + 0.03052 + 0.09156 + 0.17004 = 0.2969$$

$$P(\text{insufficient}) = P(X \geq 4) = 1 - 0.2969 \approx 0.7031$$

ANSWER

$P(\text{lane insufficient}) \approx 0.703$ (70.3%)

Q4 Ice-cream sales ~ Poisson($\lambda=10$ /hour). (a) P(at least 1 in first hour)? (b) Stock needed to be 95% sure?

DISTRIBUTION – POISSON $P_0(\lambda=10)$

Why Poisson?

- ✓ Sales are events in a time interval (1 hour)
- ✓ Average rate $\lambda = 10/\text{hour}$ is given — no fixed "n" trials
- ✓ Sales are independent of each other
- ✓ Two sales can't happen at exactly the same instant
- $X \sim \text{Po}(10)$

PART (A) — $P(X \geq 1)$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{10^0 e^{-10}}{0!} = 1 - e^{-10} \approx 1 - 0.0000454 \approx 0.99995$$

PART (B) — FIND STOCK S SUCH THAT $P(X \leq S) \geq 0.95$

We need the smallest s where the cumulative Poisson probability is ≥ 0.95 :

S	$P(X \leq S)$ APPROX.	$\geq 0.95?$
10	≈ 0.5831	No
13	≈ 0.8645	No
15	≈ 0.9513	✓ Yes
14	≈ 0.9165	No

ANSWERS

(a) $P(X \geq 1) \approx 0.99995$ · (b) Stock at least 15 ice-creams

Q5 Pump fails on average once per 500 hours. P(fails exactly 3 times in 2000 hours)?

DISTRIBUTION — POISSON $\text{PO}(\lambda=4)$

Why Poisson? And why $\lambda=4$?

Failures are rare events over continuous time. Rate = 1/500 per hour. The interval is 2000 hours. Scale: $\lambda = (1/500) \times 2000 = 4$ failures expected in 2000 hours.

COMPUTE $P(X=3)$ WITH $\lambda=4$

$$P(X = 3) = \frac{4^3 e^{-4}}{3!} = \frac{64 \times e^{-4}}{6} = \frac{64 \times 0.01832}{6} = \frac{1.1725}{6} \approx 0.1954$$

ANSWER

$P(X=3) \approx 0.1954$ (about 19.5%)

Problem Sheet 9 — All Discrete Distributions

This sheet mixes Binomial, Poisson, Geometric, Negative Binomial and Hypergeometric. Identification is the key skill.

Q2 Average demand = 8 motors/week. Delivery takes 1 week. How low can stock fall and still be 95% sure of meeting demand?

DISTRIBUTION — POISSON $\text{PO}(\lambda=8)$

Why Poisson?

Demand for motors = count of a rare-ish event per week. No fixed number of customers. Average rate $\lambda = 8/\text{week}$. Events (orders) are independent. $\rightarrow X \sim \text{Po}(8)$.

FIND MINIMUM STOCK S SUCH THAT $P(X \leq S) \geq 0.95$

S	$P(X \leq S)$ FOR $\text{PO}(8)$	$\geq 0.95?$
8	≈ 0.5926	No
10	≈ 0.8159	No
12	≈ 0.9362	No
13	≈ 0.9658	✓ Yes

ANSWER

Stock must not fall below 13 motors before ordering.

Q3 Die tossed until 6 appears. (a) $P(6 \text{ for first time after 10 tosses})$. (b) $P(6 \text{ for 6th time after 10 tosses})$.

PART (A) – GEOMETRIC $\text{Geo}(p=1/6)$

Why Geometric?

"First time" a 6 appears = first success. Trials are independent. $p = 1/6$ each roll. $X = \text{trial number of first 6}$. $\rightarrow X \sim \text{Geo}(1/6)$. "Before 6 appears" in first 10 tosses = 6 does NOT appear in first 10 = $P(X > 10)$.

PART (A): $P(\text{FIRST 6 APPEARS AFTER TOSS 10}) = P(X > 10)$

$$P(X > 10) = (1 - p)^{10} = \left(\frac{5}{6}\right)^{10} \approx 0.1615$$

PART (B) – NEGATIVE BINOMIAL $\text{NB}(r=6, p=1/6)$

Why Negative Binomial?

"6th time" a 6 appears = $r=6$ successes. Target r successes. $P(r\text{-th success has NOT occurred in first 10 trials}) = P(\text{fewer than 6 sixes in 10 rolls})$. This is equivalent to Binomial $B(10, 1/6)$: $P(X < 6)$.

PART (B): $P(\text{FEWER THAN 6 SIXES IN 10 ROLLS}) = P(X < 6) \sim B(10, 1/6)$

$$P(X < 6) = \sum_{k=0}^5 \binom{10}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{10-k} \approx 0.9999$$

(All values close to 1 since getting 6 sixes in 10 rolls with $p=1/6$ is extremely rare)

ANSWERS

(a) $P(X > 10) = (5/6)^{10} \approx 0.1615$ · (b) ≈ 0.9999

Q4 Presidential candidate needs 50 signatures. $p=0.15$ per delegate. $P(\text{need 300 delegates})?$

DISTRIBUTION – NEGATIVE BINOMIAL $\text{NB}(r=50, p=0.15)$

Why Negative Binomial?

She needs exactly $r=50$ signatures (successes). She approaches delegates one at a time — each independently gives signature ($p=0.15$) or not. She stops at the 50th signature. $\rightarrow X = \text{total delegates approached} = \text{NB}(50, 0.15)$.

Question: $P(X = 300)$ — 50th signature comes from the 300th delegate.

$$P(X = 300) = \binom{299}{49} (0.15)^{50} (0.85)^{250}$$

This is extremely small numerically. In practice we note:

$$\text{Mean} = \frac{r}{p} = \frac{50}{0.15} \approx 333.3 \quad \text{SD} = \sqrt{\frac{r(1-p)}{p^2}} \approx 44.6$$

300 is about $(333-300)/44.6 \approx 0.74$ SD below the mean — so it's a moderate probability region but the exact value requires computation.

ANSWER

$P(X=300) = C(299,49) \cdot (0.15)^{50} \cdot (0.85)^{250}$ (evaluate numerically; 300 is near the mean of ~333)

Q5 10 cartons: 3 sour, 7 fresh. You select 4 at random. $P(\text{at least half are sour}) = P(X \geq 2)$?

DISTRIBUTION - HYPERGEOMETRIC $H(N=10, K=3, n=4)$

Why Hypergeometric and NOT Binomial?

You're selecting from a FINITE population of 10 cartons WITHOUT replacement. Once you pick a carton it's gone — so the probability of getting a sour one changes each draw. This violates the Binomial's independence requirement. The population is small ($N=10$), so the finite population correction matters enormously.

✓ $N=10$ (population size) · ✓ $K=3$ (sour = successes in population) · ✓ $n=4$ (sample size) · ✓ WITHOUT replacement

STEP 1 - UNDERSTAND THE FORMULA

$$P(X = k) = \frac{\binom{3}{k} \binom{7}{4-k}}{\binom{10}{4}}, \quad k = 0, 1, 2, 3$$

$$\binom{10}{4} = 210$$

STEP 2 - COMPUTE $P(X=0)$, $P(X=1)$

$$P(X = 0) = \frac{\binom{3}{0} \binom{7}{4}}{210} = \frac{1 \times 35}{210} = \frac{35}{210} \approx 0.1667$$

$$P(X = 1) = \frac{\binom{3}{1} \binom{7}{3}}{210} = \frac{3 \times 35}{210} = \frac{105}{210} = 0.5$$

STEP 3 - $P(X \geq 2) = 1 - P(X=0) - P(X=1)$

$$P(X \geq 2) = 1 - \frac{35}{210} - \frac{105}{210} = 1 - \frac{140}{210} = \frac{70}{210} = \frac{1}{3} \approx 0.333$$

Verification: $P(X=2) = C(3,2) \times C(7,2)/210 = 3 \times 21/210 = 63/210 = 0.3$ and $P(X=3) = C(3,3) \times C(7,1)/210 = 7/210 = 0.033$. Sum = $0.3 + 0.033 = 0.333$ ✓

ANSWER

$$P(\text{at least 2 sour}) = 1/3 \approx 0.333$$

Problem Sheet 10 — Continuous Distributions

Uniform, Exponential and Normal distributions. Continuous RVs — all probabilities are areas under curves.

Q1 Radio announces time every 15 minutes. You wake at random. P(wait less than 5 minutes)?

DISTRIBUTION — UNIFORM $U(0, 15)$

Why Uniform?

You wake at a completely random time within the 15-minute cycle. There is no reason to think any moment is more likely than any other — all moments are equally likely. X = minutes since last announcement $\sim U(0, 15)$. You wait $15-X$ minutes for the next announcement. You want $P(15-X < 5) = P(X > 10)$.

UNIFORM PDF AND PROBABILITY

$$f(x) = \frac{1}{15}, \quad 0 \leq x \leq 15$$

$$P(\text{wait} < 5) = P(X > 10) = \int_{10}^{15} \frac{1}{15} dx = \frac{5}{15} = \frac{1}{3} \approx 0.333$$

ANSWER

$$P(\text{wait} < 5 \text{ min}) = 1/3 \approx 0.333$$

Q2 Device lifetime $\sim \text{Exp}(\lambda=1/1000)$. (a) P(lasts <1000h)? (b) P(lasts >1200h)? (c) If three devices, P(one lasts <800h)?

DISTRIBUTION — EXPONENTIAL $\text{EXP}(\lambda=1/1000)$

Why Exponential?

Lifetime/failure time of a device is a continuous waiting time. "Exponential distribution with parameter $\lambda=1/1000$ " is stated directly. Mean = $1/\lambda = 1000$ hours. CDF: $F(t) = 1 - e^{-t/1000}$.

PART (A) — $P(X < 1000)$

$$P(X < 1000) = 1 - e^{-1000/1000} = 1 - e^{-1} \approx 1 - 0.3679 = 0.6321$$

PART (B) — $P(X > 1200) = 1 - P(X \leq 1200)$

$$P(X > 1200) = e^{-1200/1000} = e^{-1.2} \approx 0.3012$$

PART (C) — $P(\text{EXACTLY 1 OF 3 DEVICES LASTS } < 800\text{H})$

First find $p = P(\text{one device lasts} < 800\text{h})$:

$$p = P(X < 800) = 1 - e^{-0.8} \approx 1 - 0.4493 = 0.5507$$

Now 3 devices, want exactly 1 lasts < 800h → **Binomial B(3, 0.5507)**:

$$P(Y = 1) = \binom{3}{1} (0.5507)^1 (0.4493)^2 = 3(0.5507)(0.2019) \approx 0.3334$$

ANSWERS

(a) 0.6321 · (b) 0.3012 · (c) ≈ 0.333

Q3 Customers arrive at 30/hour. 1 minute has passed since last. P(at least 1 more minute before next)?

DISTRIBUTION – EXPONENTIAL (MEMORYLESS PROPERTY)

Why Exponential + Memoryless?

Customers arrive at Poisson rate 30/hour = 0.5/minute. Inter-arrival time $X \sim \text{Exp}(\lambda=0.5)$. The question uses conditional probability $P(X > 2 \mid X > 1)$. By the MEMORYLESS property of the exponential: $P(X > s+t \mid X > s) = P(X > t)$. So $P(X > 2 \mid X > 1) = P(X > 1)$ – the past minute of waiting is irrelevant.

APPLY MEMORYLESS PROPERTY

$$P(X > 2 \mid X > 1) = P(X > 1) = e^{-0.5 \times 1} = e^{-0.5} \approx 0.6065$$

ANSWER

P(wait ≥ 1 more minute) = $e^{-0.5}$ ≈ 0.6065

Q4 $X \sim N(\mu=3, \sigma^2=4)$. Find (a) $P(X < 4)$, (b) $P(|X| < 6)$, (c) a such that $P(X < a) = 0.80$.

DISTRIBUTION – NORMAL $N(\mu=3, \Sigma^2=4)$, SO $\Sigma=2$

Why Normal?

Stated directly: "X is distributed as a normal variable with mean $\mu=3$ and variance $\sigma^2=4$ ". $\sigma = \sqrt{4} = 2$. Convert to z-scores to use standard normal table.

PART (A) – $P(X < 4)$

$$z = \frac{4-3}{2} = 0.5 \implies P(X < 4) = P(Z < 0.5) = \Phi(0.5) \approx 0.6915$$

PART (B) – $P(|X| < 6) = P(-6 < X < 6)$

$$z_1 = \frac{-6-3}{2} = -4.5 \quad z_2 = \frac{6-3}{2} = 1.5$$

$\backslash [P(-4.5$

PART (C) – FIND A: $P(X < A) = 0.80$

From tables: $\Phi(z) = 0.80 \rightarrow z \approx 0.8416$

$$z = \frac{a-\mu}{\sigma} \implies a = \mu + z\sigma = 3 + 0.8416 \times 2 = 3 + 1.683 = 4.683$$

ANSWERS

(a) 0.6915 · (b) ≈ 0.9332 · (c) a ≈ 4.68

Q5 $X \sim \text{Normal}$ with $\sigma^2=16$. $P(X < 10) = 0.8413$. Find μ .

REVERSE NORMAL – FIND μ FROM A GIVEN PROBABILITY

$\sigma = \sqrt{16} = 4$. $P(X < 10) = 0.8413 \rightarrow$ from tables, $\Phi(1.0) \approx 0.8413$, so $z = 1.0$.

$$z = \frac{x - \mu}{\sigma} \implies 1.0 = \frac{10 - \mu}{4} \implies 10 - \mu = 4 \implies \mu = 6$$

ANSWER

$\mu = 6$

Q6 Masses $\sim N(65\text{kg}, \sigma=15\text{kg})$. Lift limit: 600kg for 10 passengers. How often exceeded?

DISTRIBUTION – NORMAL, BUT FOR THE SUM OF 10 PEOPLE

Key insight – distribution of a SUM

Each person's mass $X_i \sim N(65, 15^2)$. The sum $S = X_1 + X_2 + \dots + X_{10}$ of n independent normal RVs is also normal. Mean of sum = $n \times \mu = 10 \times 65 = 650\text{kg}$. Variance of sum = $n \times \sigma^2 = 10 \times 225 = 2250$. SD of sum = $\sqrt{2250} \approx 47.43\text{kg}$.

SET UP: $S \sim N(650, 2250)$. FIND $P(S > 600)$.

$$z = \frac{600 - 650}{\sqrt{2250}} = \frac{-50}{47.43} \approx -1.054$$

$$P(S > 600) = P(Z > -1.054) = 1 - P(Z < -1.054) \approx 1 - 0.1459 = 0.8541$$

ANSWER

The 600kg limit is exceeded approximately 85.4% of the time. The lift is dangerously overloaded on average!